TURBULENCE CLOSURE: RANS, LES, AND HYBRID IDEAS

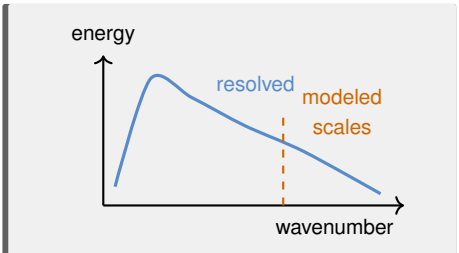
What is averaged, what is filtered, and what must be modeled?

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Fluid Dynamics and IBM Theory • Module 5

Adapted in part from lectures by Steven L. Brunton; equations follow Pope (2000) and Wilcox (2006).



Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

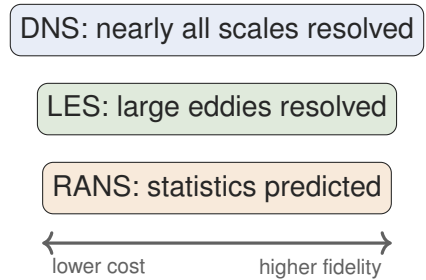
WHY TURBULENCE MODELING?

Engineering design asks for quantities such as

- lift, drag, stall, and wall loading;
- wake recovery, mixing, heat transfer, and noise;
- performance over *many* geometries and operating points.

The modeling bargain

Spend fewer degrees of freedom on the smallest motions, but retain the physics needed for the chosen quantity of interest.



No method is “best” without a flow, grid, wall treatment, and target output.

DNS RESOLVES THE CASCADE—AT A STEEP COST

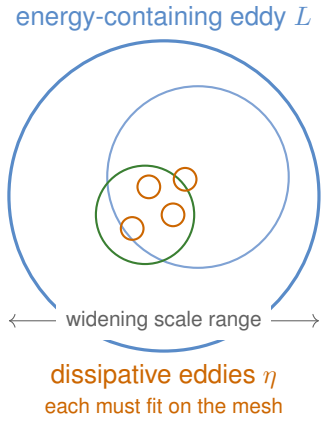
DNS advances the Navier–Stokes equations without a turbulence closure and resolves down to the dissipative scales.
For idealized three-dimensional turbulence,

$$\frac{\eta}{L} \sim \mathrm{Re}_L^{-3/4}, \qquad N_{\mathrm{grid}} \sim \left(\frac{L}{\eta}\right)^3 \sim \mathrm{Re}_L^{9/4}.$$

The time-step count also grows as the smallest time scale shrinks.

Interpret the scaling carefully

It is an order-of-magnitude argument, not a universal cost law for every geometry or numerical method.



CHOOSE THE FIDELITY THAT ANSWERS THE QUESTION

Method	Directly computed	Modeled	Typical role
DNS	all dynamically relevant velocity scales	no turbulence closure	canonical flows; reference data
LES	filtered, unsteady large scales	subfilter stress; often wall effects	wakes, mixing, separated unsteadiness
RANS/URANS	mean or phase-resolved mean field	all turbulent correlations	design sweeps; attached wall flows
Hybrid RANS–LES	RANS region plus resolved eddies elsewhere	region- and grid-dependent	high-Re separated flows

> Closure means representing terms created by averaging or filtering that are not determined by the retained variables.

RANS BEGINS WITH AN AVERAGING OPERATOR

For velocity and kinematic pressure,

$$u_i(\boldsymbol{x}, t) = U_i(\boldsymbol{x}, t) + u_i'(\boldsymbol{x}, t), \qquad p = P + p', \qquad \overline{u_i'} = \overline{p'} = 0.$$

Here $U_i = \overline{u_i}$ may be an ensemble, long-time, or statistically equivalent average.

Steady RANS

For statistically stationary flow, a long-time mean is independent of time: $\partial_t U_i = 0$.

URANS

Retain $\partial_t U_i$ when the chosen mean is time dependent; modeled turbulence still lies beneath that mean.

For incompressible constant-density flow, $\partial_i U_i = 0$ and $\partial_i u_i' = 0$.

AVERAGING CREATES THE REYNOLDS-STRESS DIVERGENCE

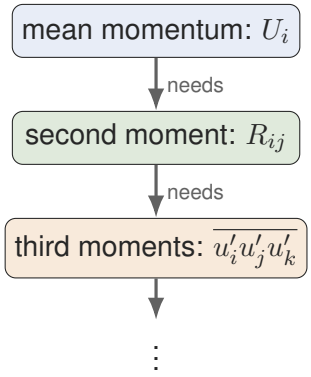
With pressure normalized by density, the incompressible RANS/URANS equations are

$$\partial_t U_i + U_j \partial_j U_i = -\partial_i P + \nu \partial_{jj} U_i - \partial_j R_{ij},$$
$$\partial_i U_i = 0, \qquad R_{ij} := \overline{u_i' u_j'}.$$

> The physical turbulent stress is $-\rho R_{ij}$; the equation contains $-\partial_j R_{ij}$. The sign convention must remain paired with the closure.

R_{ij} is symmetric and positive semidefinite. $R_{ii} = 2k$ fixes its trace through the TKE.

WHY NOT DERIVE AN EXACT EQUATION FOR R_{ij}?



Exact transport adds new unknowns:

pressure–strain correlations,
triple-velocity transport,
dissipation tensors, . . .

Closure is structural, not a bookkeeping nuisance

Moment equations form a hierarchy: an assumption, modeled equation, or data-driven relation is needed to stop it.

TURBULENT KINETIC ENERGY CONDENSES THE NORMAL STRESSES

$$k := \frac{1}{2} R_{ii} = \frac{1}{2} \left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2} \right).$$

What k measures

Fluctuation kinetic energy per unit mass. It is nonnegative and has units L^2/T^2.

What k does not determine

The scalar k fixes only $\mathrm{tr}(R)$; it does not determine stress anisotropy or shear stresses.

Modeling lens: follow how mean-flow energy becomes turbulent energy, moves through space and scale, and is dissipated by viscosity.

THE EXACT TKE BUDGET NAMES THE COMPETING PROCESSES

For incompressible, constant-property flow,

$$\partial_t k + U_j \partial_j k = \underbrace{-R_{ij}\partial_j U_i}_{\mathcal{P}} - \underbrace{\varepsilon}_{\geq 0} - \partial_j T_j,$$

$$\varepsilon = \nu \overline{\partial_j u_i' \, \partial_j u_i'}, \qquad T_j = \tfrac{1}{2}\overline{u_i' u_i' u_j'} + \overline{p' u_j'} - \nu \partial_j k.$$

This is the pseudo-dissipation; it differs from $2\nu \overline{s_{ij}' s_{ij}'}$ by a term that vanishes in homogeneous turbulence.



PRODUCTION IS A SIGNED EXCHANGE WITH THE MEAN FLOW

Because R_{ij} is symmetric,

$$\mathcal{P} = -R_{ij}\partial_j U_i = -R_{ij}S_{ij}, \qquad S_{ij} = \tfrac{1}{2}(\partial_j U_i + \partial_i U_j).$$

For a parallel shear flow $U = (U(y), 0, 0)$,

$$\mathcal{P} = -\overline{u'v'}\,\frac{\mathrm{d}U}{\mathrm{d}y}.$$

Common wall-flow signs

$\mathrm{d}U/\mathrm{d}y > 0$ and $\overline{u'v'} < 0$ give $\mathcal{P} > 0$: mean energy feeds turbulence.

Do not infer the sign from “separation”

Separated shear often produces turbulence strongly, but local production can be negative.

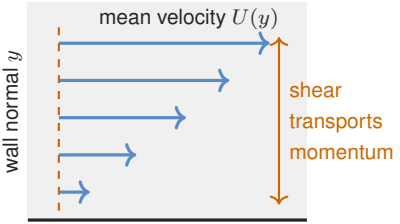
BOUSSINESQ CLOSES STRESS WITH A SCALAR EDDY VISCOSITY

The linear eddy-viscosity hypothesis is

$$-R_{ij} = 2\nu_t S_{ij} - \frac{2}{3}k\delta_{ij},$$

or equivalently $R_{ij} = \frac{2}{3}k\delta_{ij} - 2\nu_t S_{ij}$.
For incompressible flow, absorb the isotropic part into

$$P^* = P + \frac{2}{3}k.$$



Modeling task

Specify the scalar field $\nu_t \geq 0$ from algebra, auxiliary transport equations, or data.

ZERO-EQUATION BASELINE: CONSTANT ν_t

The simplest choice is

$$\nu_t = \text{constant}, \qquad \nu_{\text{eff}} = \nu + \nu_t.$$

Why keep this model?

It exposes the mechanism, is inexpensive, and can be calibrated for a narrow configuration.

Why it is not predictive

One constant cannot represent wall damping, free-stream intermittency, spatially varying shear, or history effects.

Useful baseline question: which observed behavior can be explained by enhanced momentum diffusion alone?

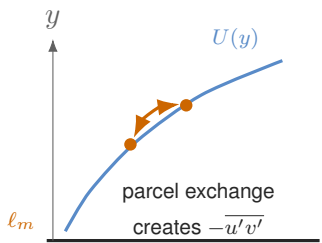
PRANDTL MIXING LENGTH CONNECTS SHEAR AND MOMENTUM EXCHANGE

For a parallel shear flow,

$$\nu_t = \ell_m^2 \left| \frac{\mathrm{d}U}{\mathrm{d}y} \right|.$$

The mixing length ℓ_m represents a displacement over which a fluid parcel retains velocity contrast.

- In the logarithmic layer, $\ell_m \approx \kappa y$ with the von Kármán constant $\kappa \approx 0.41$.
- Near the wall and outside that layer, damping and outer-length caps are needed.
- This algebraic form is not a general three-dimensional closure.



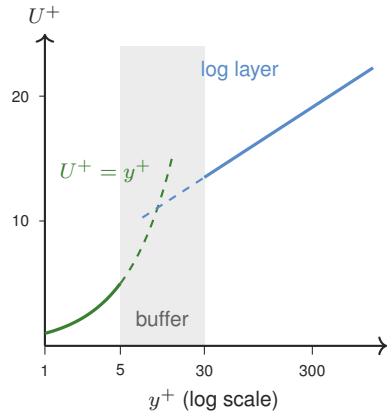
WALL SCALING AND THE LOG LAW

Wall units are built from the friction velocity:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}, \qquad y^+ = \frac{y\,u_\tau}{\nu}, \qquad U^+ = \frac{U}{u_\tau}.$$

- Viscous sublayer: $U^+ = y^+$ for $y^+ \lesssim 5$.
- Log layer: $U^+ = \dfrac{1}{\kappa}\ln y^+ + B$ with $\kappa \approx 0.41$ and $B \approx 5.0$–5.2, for $30 \lesssim y^+ \lesssim 0.1$–$0.2\,\delta^+$.
- Buffer layer ($5 \lesssim y^+ \lesssim 30$): neither law holds; turbulence production peaks near $y^+ \approx 12$.

Here $\delta^+ = \delta u_\tau/\nu$ is the friction Reynolds number; Pope (2000, Table 7.1) quotes the log law for $y^+ > 30$, $y/\delta < 0.3$, and recent high-Re data place its upper end nearer $0.15\,\delta$.



Drawn to scale ($\kappa = 0.41$, $B = 5.2$). Schematic: instructor's course notes, after Pope (2000), Ch. 7.

WALL SCALING IN PRACTICE: WALL FUNCTIONS OR WALL RESOLUTION

A wall function imposes this profile at the first cell instead of resolving the sublayer, and needs the first cell centre near $30 < y^+ < 300$. A low-Re model instead resolves to $y^+ \approx 1$.

- u_τ is part of the solution, so a wall function solves the log law at the first cell for u_τ (and hence τ_w) iteratively.
- Mesh design runs the other way: estimate u_τ from a skin-friction correlation, then place the first cell at $y = y^+\nu/u_\tau$.
- Worked example (water, $\nu = 10^{-6}\,\mathrm{m}^2/\mathrm{s}$, $u_\tau = 0.05$ m/s): $\nu/u_\tau = 20\,\mu$m, so $y^+ = 1$ is $y = 20\,\mu$m and $y^+ = 30$ is $y = 0.6$ mm.
- A first cell in the buffer layer ($5 \lesssim y^+ \lesssim 30$) satisfies neither approach; many codes then blend the two laws.

TWO-EQUATION MODELS EVOLVE TURBULENCE SCALES

Family	Transported variables	Eddy viscosity	Key sensitivity
standard k–ε	TKE k and dissipation rate ε	$\nu_t = C_\mu k^2/\varepsilon$	near-wall treatment; separation
k–ω	TKE k and specific dissipation ω	$\nu_t = k/\omega$ (Wilcox)	free-stream ω; practical limiters
blended variants, e.g. Menter's SST k–ω (blending functions F_1, F_2 and a shear-stress limiter)	combine near-wall and outer-flow behavior	formulation dependent	calibration and blending functions

For the *standard* k–ε model, $C_\mu \approx 0.09$ is a conventional empirical value; it is not universal across all variants. The standard constants are $C_\mu = 0.09$, $C_{\varepsilon 1} = 1.44$, $C_{\varepsilon 2} = 1.92$, $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.3$ (Launder & Spalding 1974).

> Extra PDEs introduce memory, transport, and length/time scales—but they do not remove modeling assumptions.

THE k–ε AND k–ω TRANSPORT EQUATIONS

Standard k–ε (Launder & Spalding 1974)

$$\partial_t k + U_j \partial_j k = P_k - \varepsilon + \partial_j \left[\left(\nu + \tfrac{\nu_t}{\sigma_k} \right) \partial_j k \right],$$

$$\partial_t \varepsilon + U_j \partial_j \varepsilon = C_{\varepsilon 1} \frac{\varepsilon}{k} P_k - C_{\varepsilon 2} \frac{\varepsilon^2}{k} + \partial_j \left[\left(\nu + \tfrac{\nu_t}{\sigma_\varepsilon} \right) \partial_j \varepsilon \right],$$

$$\nu_t = C_\mu \frac{k^2}{\varepsilon}.$$

$$P_k = -\overline{u_i' u_j'}\, \partial_j U_i = 2 \nu_t S_{ij} S_{ij}.$$

Constants: $C_\mu = 0.09$, $C_{\varepsilon 1} = 1.44$, $C_{\varepsilon 2} = 1.92$, $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.3$.

Wilcox k–ω (1988 form; the 2006 revision adds a cross-diffusion term and a stress limiter on ν_t, and recalibrates the constants)

$$\partial_t k + U_j \partial_j k = P_k - \beta^* k \omega + \partial_j \left[(\nu + \sigma_k \nu_t)\, \partial_j k \right],$$

$$\partial_t \omega + U_j \partial_j \omega = \alpha \frac{\omega}{k} P_k - \beta \omega^2 + \partial_j \left[(\nu + \sigma_\omega \nu_t)\, \partial_j \omega \right],$$

$$\nu_t = \frac{k}{\omega}.$$

1988 constants: $\alpha = 5/9$, $\beta = 3/40$, $\beta^* = 0.09$, $\sigma_k = \sigma_\omega = 0.5$.
Note $\varepsilon \approx \beta^* k \omega$, so ω is a specific dissipation rate. (Wilcox defines $\omega = \varepsilon / (\beta^* k)$, so this holds by definition; diffusion coefficients multiply ν_t here but divide it in k–ε.)

NEAR-WALL BEHAVIOUR: WHY SST BLENDS THE TWO MODELS

> k–ε needs wall functions or damping terms near walls, because ε does not behave well as $y \to 0$ (at the wall $k \to 0$ like y^2 while ε stays finite, so the destruction term $C_{\varepsilon 2}\varepsilon^2/k$ is singular and the model's near-wall ν_t is wrong without damping functions).
>
> k–ω integrates cleanly to the wall but is sensitive to the free-stream value of ω—which is exactly why Menter's SST blends k–ω near the wall with k–ε in the outer flow.

- Menter (1994) rewrites k–ε in terms of ω; the result is the k–ω equations plus a cross-diffusion term $2(1 - F_1)\sigma_{\omega 2}\, \omega^{-1} \partial_j k\, \partial_j \omega$.
- The blending function F_1 is ≈ 1 near the wall (k–ω) and ≈ 0 in the outer layer and free stream (k–ε behavior).
- The SST limiter $\nu_t = a_1 k / \max(a_1 \omega,\, \Omega F_2)$, $a_1 = 0.31$ (Ω: vorticity magnitude; later versions use the strain rate S), caps the shear stress at $\approx a_1 k$ (Bradshaw's assumption), improving adverse-pressure-gradient predictions.

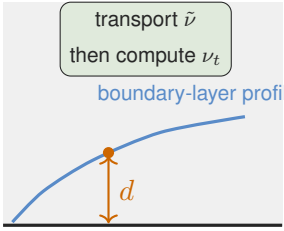
SPALART–ALLMARAS: A ONE-EQUATION AERODYNAMIC MODEL

SA transports a modified kinematic viscosity $\tilde{\nu}$ and reconstructs

$$\nu_t = \tilde{\nu} f_{v1}.$$

Its transport equation balances production, diffusion, and near-wall destruction, with wall distance d entering the model.

- designed for aerodynamic, predominantly attached boundary layers;
- relatively economical and extensively used;
- behavior in separation, transition, rotation, and free shear depends on variant and setup.



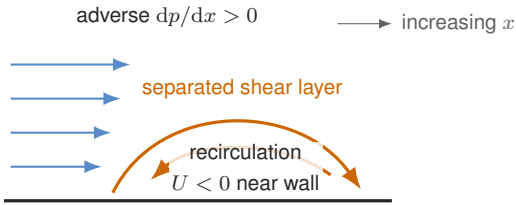
Model definition and variants: NASA Turbulence Modeling Resource.

SEPARATION EXPOSES LIMITS OF SCALAR EDDY VISCOSITY

Adverse pressure gradients can decelerate near-wall fluid, producing reversal and a recirculation region.
Linear eddy-viscosity models constrain stress anisotropy to align with the mean strain tensor. They may miss

- anisotropy and curvature/rotation effects,
- nonequilibrium response and history,
- the onset and extent of separation.

This is a limitation of the closure class, not of Reynolds averaging itself.



RANS CHECKPOINT: KEEP THREE LAYERS DISTINCT

1. Exact averaged equation: $R_{ij} = \overline{u_i' u_j'}$ appears through $-\partial_j R_{ij}$.
2. Constitutive hypothesis: a scalar eddy viscosity sets $-R_{ij}^{\mathrm{dev}} = 2\nu_t S_{ij}$.
3. Model for ν_t: constant, mixing length, k–ε, k–ω, SA, or another closure.

> Changing the equation for ν_t cannot repair every limitation imposed by the scalar Boussinesq relation.

LES ASKS A DIFFERENT QUESTION

RANS asks: *what are the statistics?*

LES asks: *which motions can this grid resolve, and how do the rest affect them?*

Resolve

Large, geometry-dependent, energy-containing motions and their unsteady transport.

Model

The effect of motions below the filter width, including energy transfer across the cutoff.

Conditional advantage

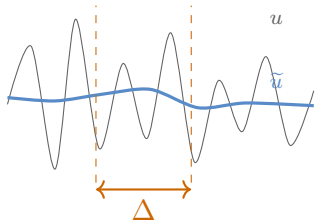
LES can improve predictions of wakes and separated unsteadiness only when grid resolution, numerics, inflow, SGS closure, and wall treatment are adequate.

FILTERING IS NOT REYNOLDS AVERAGING

Define a spatially filtered field by convolution,

$$\widetilde{u}_i(\boldsymbol{x}, t) = \int G_\Delta(\boldsymbol{x} - \boldsymbol{r}) u_i(\boldsymbol{r}, t)\, \mathrm{d}\boldsymbol{r}.$$

- $\widetilde{u}_i$ is a coarse-grained field, not generally an ensemble mean.
- A finite-width filter is generally not idempotent: $\widetilde{\widetilde{u}} \neq \widetilde{u}$.
- In implicit LES, grid and numerics help define the effective filter.



FILTERING THE NONLINEARITY CREATES SGS STRESS

Assuming a homogeneous filter that commutes with differentiation,

$$\partial_t \widetilde{u}_i + \widetilde{u}_j \partial_j \widetilde{u}_i = -\partial_i \widetilde{p} + \nu \partial_{jj} \widetilde{u}_i - \partial_j \tau_{ij},$$
$$\partial_i \widetilde{u}_i = 0,$$

with

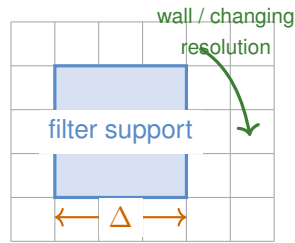
$$\boxed{\tau_{ij} := \widetilde{u_i u_j} - \widetilde{u}_i \widetilde{u}_j.}$$

> There is no need to assert a unique additive “subgrid velocity.” The exact closure term is the difference between a filtered product and the product of filtered fields.

THE LES CUTOFF IS TIED TO GRID AND NUMERICS

In an explicit-filter view, G_Δ defines the cutoff. In many practical LES calculations, the discretization supplies an *effective* filter.

- Variable filter width and boundaries can make filtering and differentiation fail to commute.
- A nominal SGS model cannot compensate for an under-resolved, overly dissipative, or dispersive scheme in every regime.
- Report Δ, mesh anisotropy, and numerical method with the closure.



SMAGORINSKY CLOSES THE DEVIATORIC SGS STRESS

The eddy-viscosity SGS hypothesis is

$$\tau_{ij}^{\mathrm{dev}} = -2\nu_{\mathrm{sgs}}\widetilde{S}_{ij}, \qquad \widetilde{S}_{ij} = \tfrac{1}{2}(\partial_j \widetilde{u}_i + \partial_i \widetilde{u}_j).$$

The classical Smagorinsky model sets

$$\nu_{\mathrm{sgs}} = (C_s\Delta)^2|\widetilde{S}|, \qquad |\widetilde{S}| = \sqrt{2\widetilde{S}_{ij}\widetilde{S}_{ij}}.$$

$C_s \approx 0.17$ for isotropic turbulence (Lilly), reduced to ≈ 0.1 in shear flows. (Lilly 1967 obtained $C_s \approx 0.17$ by matching the mean SGS dissipation to a Kolmogorov inertial range with $C_K \approx 1.5$; Deardorff 1970 found ≈ 0.1 necessary in channel flow, where mean shear adds to $|\widetilde{S}|$.)

For $\nu_{\mathrm{sgs}} \geq 0$, the modeled transfer is

$$\Pi = -\tau_{ij}\widetilde{S}_{ij} = 2\nu_{\mathrm{sgs}}\widetilde{S}_{ij}\widetilde{S}_{ij} \geq 0.$$

The isotropic part of τ_{ij} drops out because $\widetilde{S}_{ii} = 0$.

Consequence

The basic model is purely dissipative and needs wall damping or a dynamic treatment near walls.

THE GERMANO PROCEDURE ESTIMATES COEFFICIENTS FROM RESOLVED SCALES

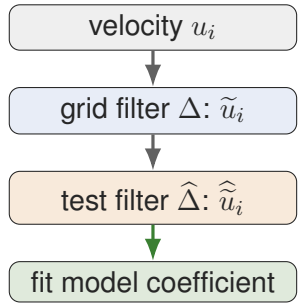
Apply a second, wider *test filter* $\widehat{(\cdot)}$ and use the identity

$$L_{ij} = T_{ij} - \widehat{\tau}_{ij} = \widehat{\widetilde{u}_i \widetilde{u}_j} - \widehat{\widetilde{u}}_i \widehat{\widetilde{u}}_j.$$

Here $T_{ij} := \widehat{\widetilde{u_i u_j}} - \widehat{\widetilde{u}}_i \widehat{\widetilde{u}}_j$ is the stress at the combined (grid then test) filter level. L_{ij} is computable from the resolved field alone.

A modeled form on both filter levels gives an equation for a local or averaged coefficient.

- adapts to flow and resolution;
- clipping/averaging is often used for robustness;
- local backscatter can occur when the inferred coefficient is negative.

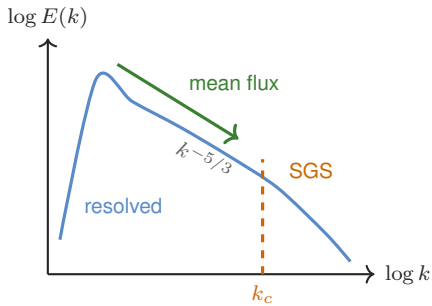


THE SPECTRAL PICTURE MOTIVATES—BUT DOES NOT PROVE—EDDY VISCOSITY

In three-dimensional turbulence, kinetic energy transfers toward smaller scales *on average* through the inertial range and is dissipated near the Kolmogorov scale.

- LES resolves $k \lesssim k_c$ (length scales $\gtrsim \Delta$); adequate numerical resolution is still required.
- SGS models represent net transfer through that cutoff.
- Instantaneous, local transfer can be upscale (backscatter).

The $-5/3$ spectrum is an inertial-range idealization, not a guarantee in every finite, anisotropic flow.



LES VERSUS RANS: COMPARE REQUIREMENTS, NOT SLOGANS

	RANS/URANS	**LES**
Primary output	mean or phase-resolved mean	filtered unsteady field plus statistics
Closure target	Reynolds stress R_{ij}	SGS stress τ_{ij}
Grid demand	resolve mean gradients and wall treatment	resolve energy-containing eddies; wall-resolved demand grows steeply with Re
Strength	economical design calculations	unsteady structures, mixing, wakes
Risk	closure bias in nonequilibrium flow	grid/numerics/inflow/wall-model dependence

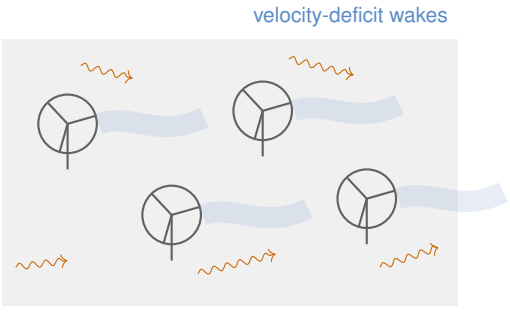
> LES is not generically “more accurate.” Its advantage is conditional on resolving the motions that control the target quantity.

APPLICATION: LES CAN EXPOSE INTERACTING TURBINE WAKES

Wind-farm calculations illustrate why resolved unsteadiness matters:

- atmospheric turbulence enters the array;
- rotating turbines extract momentum;
- wakes expand, meander, and merge;
- downstream power and loading depend on recovery.

The turbine and near-wall physics may still require actuator and wall models.

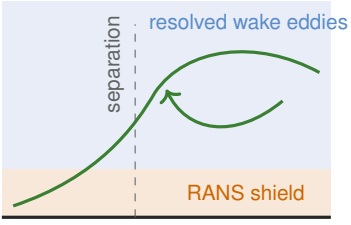


HYBRID RANS–LES PARTITIONS MODELED AND RESOLVED REGIONS

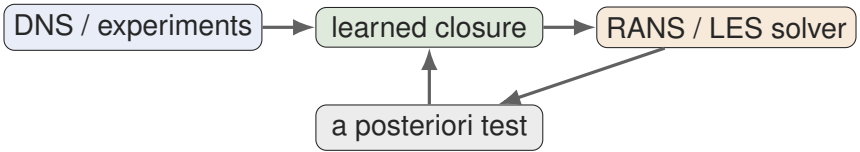
Detached-Eddy Simulation (DES) was introduced as an SA-based method that uses a grid-sensitive length scale:

- RANS-like behavior in attached near-wall layers;
- LES-like behavior where the grid supports separated eddies.

It is not a sharp switch based only on “steady” versus “unsteady” flow. Grid-induced separation and modeled-stress depletion motivated shielding strategies such as DDES.



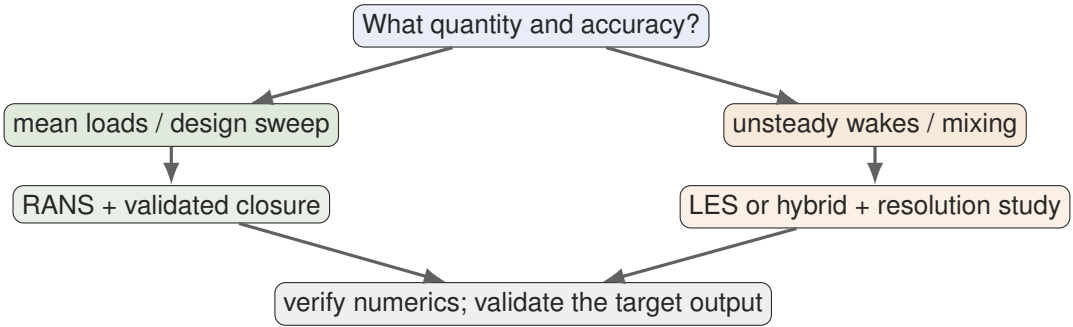
MACHINE-LEARNED CLOSURE MUST SURVIVE THE SOLVER LOOP



Useful training accuracy is not sufficient. A closure should respect relevant

- symmetries/invariance and dimensional consistency;
- realizability and appropriate energy transfer;
- numerical stability and out-of-distribution behavior;
- coupled, *a posteriori* performance in the PDE solver.

A CLOSURE DECISION TREE



RANS averages; LES filters. Both create exact unresolved stresses, and every practical closure carries assumptions that must be tested in the intended regime.

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- All schematics in this deck: instructor's course notes.